
SOC-Automation-Lab

Security Operations Center Automation Project

Project Name	SOC-Automation-Lab
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Status	Active Development

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1. Project Overview

SOC-Automation-Lab is a fully functional, home-lab based Security Operations Center (SOC) automation environment designed to simulate real-world enterprise security workflows. The project integrates industry-standard open-source tools to automate threat detection, alerting, incident response, and case management — replacing repetitive manual analyst tasks with intelligent, AI-assisted automation pipelines.

The lab topology connects a vulnerable DVWA application exposed to the internet with a centralized SIEM (Splunk), automation engine (n8n), and an AI inference layer (Gemini) — all orchestrated to respond to security events with minimal human intervention.

1.1 Objectives

- Build a realistic SOC automation pipeline using open-source tools
- Ingest security logs from Ubuntu endpoint into Splunk SIEM
- Trigger automated workflows in n8n based on Splunk alerts
- Enrich alerts with AI (Gemini) analysis for threat classification
- Send analyst notifications via Slack and Email
- Automate response actions via SSH on compromised endpoints
- Provide hands-on experience in SIEM, and blue-team operations

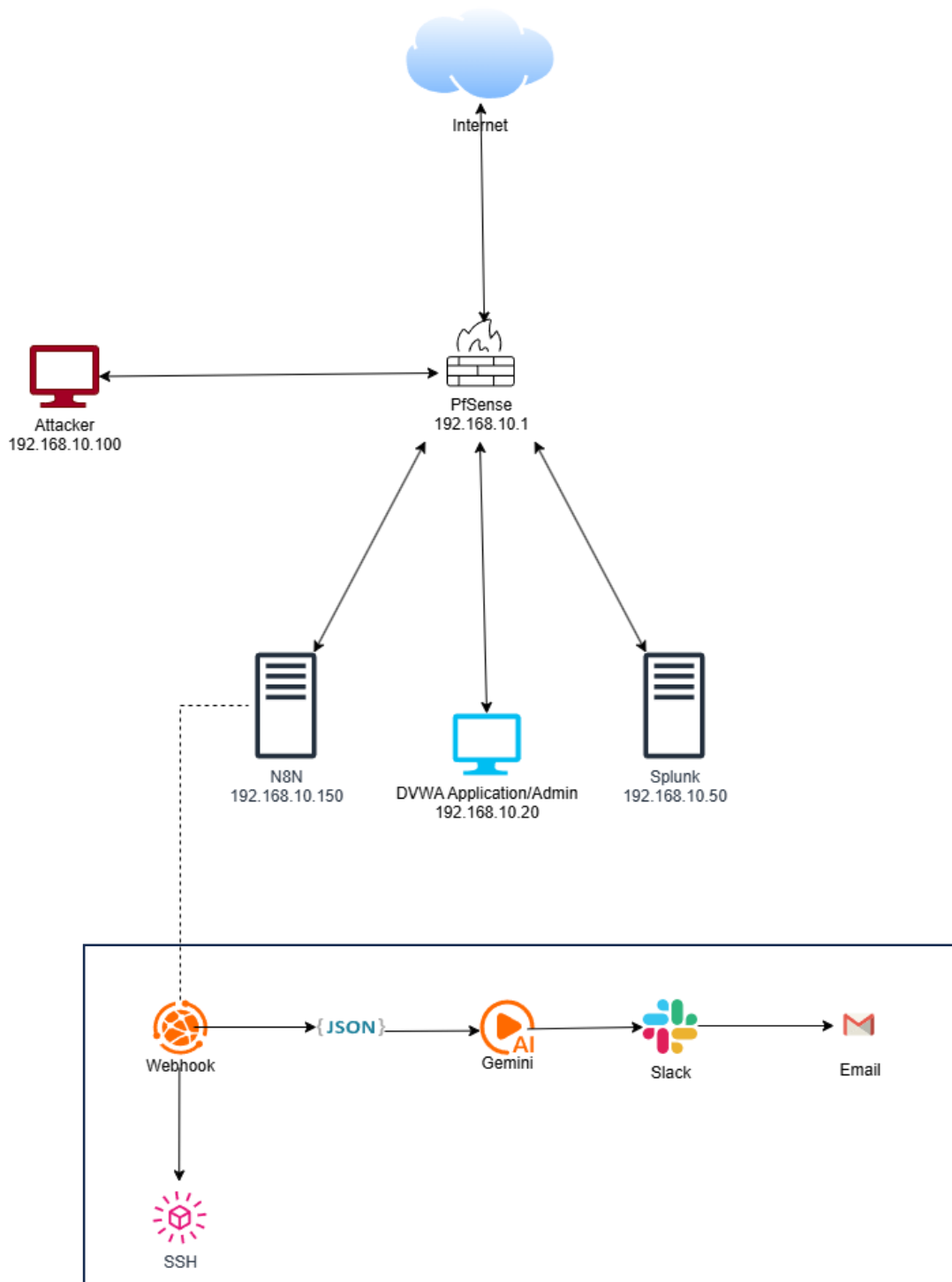
1.2 Key Technologies

Category	Purpose
Virtual Box (Ubuntu server, Ubuntu Desktop, Kali, pfsense)	to run multiple operating systems (guest OS) simultaneously within a single physical computer (host OS)
SIEM (Splunk)	Log aggregation, alerting, and threat detection
Automation (n8n)	Workflow orchestration and automation engine
pfsense	Network firewall
AI Enrichment (Gemini)	Automated threat analysis and decision support
Notification (Slack, Gmail)	Incident email notifications to SOC team
Response Action (SSH)	Remote command execution on endpoints
Draw.io	For topology creation

Note: n8n and Splunk are installed and configured on Ubuntu Server

2. Network Topology

The SOC-Automation-Lab network is built on a private subnet (192.168.10.0/24). A Ubuntu victim machine is exposed to the internet to simulate real attack scenarios. All other components reside on the internal network and communicate over defined IP addresses.



2.1 Topology Diagram Description

Based on the lab diagram, the following components and their interconnections are defined:

- Internet — External threat actor entry point connected to the lab network
- Ubuntu (Victim Machine) — Internet-facing endpoint, exposed to simulate real-world attacks; shown in blue indicating it is the compromised/monitored host
- n8n Server (192.168.10.150) — automation engine; acts as the workflow orchestrator
- Splunk Server (192.168.10.50) — SIEM for log ingestion and correlation; triggers alerts
- Integration Layer — JSON data exchange, AI enrichment, Slack, Email, and SSH response actions

2.2 Network Address Table

Component	IP Address	Role
Internet	Public / DHCP	External threat source, attack simulation
Ubuntu Victim	192.168.10.20	Internet-facing compromised endpoint
n8n Server	192.168.10.150	automation engine
Splunk Server	192.168.10.50	SIEM — log ingestion and alerting
PfSense	192.168.10.1	Gateway/Firewall
Attacker	192.168.10.100	Performing Malicious request

2.3 Data Flow

The automation data flow follows this sequence:

1. Ubuntu machine generates security events (login attempts (Brute force), SQLi)
2. Splunk Universal Forwarder ships logs to Splunk SIEM (192.168.10.50)
3. Splunk correlation rules trigger an alert on suspicious activity
4. Alert is sent via webhook to n8n automation engine (192.168.10.150)
5. n8n parses the alert payload (JSON) and routes to the automation workflow
6. AI module (Gemini) enriches the alert with threat analysis
7. Analyst is notified via Slack message and Email
8. SSH response action is executed on the endpoint to block attacker IP

3. Installation Guide

This section covers the complete installation and configuration of each component in the SOC-Automation-Lab. All installations are performed on Ubuntu Server 22.04 LTS unless otherwise specified.

3.1 VirtualBox Installation (Lab Environment Setup)

3.1.1 Download VirtualBox

<https://www.virtualbox.org/>

3.1.2 Installation Steps (Windows Host)

3.1.3 Install Extension Pack

3.1.4 Create Virtual Machines

3.1.5 Network Configuration

Adapter	Mode	Purpose
Adapter 1	NAT	Internet access
Adapter 2	Internal Network	SOC communication (192.168.10.0/24)

3.1.6 Internal Network Setup

3.1.7 Verify Virtualization

```
ping 192.168.10.1
```

3.2 Ubuntu Server Installation (SOC Core Systems)

Step 1: Download Ubuntu Server

Step 2: Create Virtual Machine

Step 3: Install Ubuntu Server

Step 4: Enable OpenSSH

Step 5: Update System

Step 6: Configure Static IP (If Needed)

3.3 Ubuntu Desktop (DVWA Victim Machine – 192.168.10.20)

Step 1: Download Ubuntu Desktop

Step 2: Create Virtual Machine

Step 3: Install OS

Step 4: Set Static IP

Edit:

```
sudo nano /etc/netplan/*.yaml
```

DVWA INSTALLATION

3.3.1 Install Apache, MySQL, PHP

3.3.2 Download DVWA

3.3.3 Configure Permissions

3.3.4 Configure DVWA

Update DB credentials:

3.3.5 Configure MySQL Database

3.3.6 Restart Services

3.3.7 Access DVWA

3.3.8 Set DVWA Security Level

3.4 pfSense Firewall & Gateway Installation

System Requirements

Installation Steps

3.4.1 Create Virtual Machine

Adapter	Mode	Purpose
Adapter 1	NAT / Bridged	WAN (Internet)
Adapter 2	Host-Only / Internal	LAN (SOC Network)

Access Web Interface

From any internal machine (Ubuntu / host):

<http://192.168.10.1>

3.5 Splunk SIEM (192.168.10.50)

System Requirements

- OS: Ubuntu Server 22.04 LTS
- RAM: Minimum 8 GB (16 GB recommended)
- CPU: 4 vCPUs
- Disk: 50 GB minimum
- Port: 8000 (Web UI), 9997 (Forwarder)

3.5.1 Splunk Enterprise Installation

Step 1: Download Splunk Enterprise from the official Splunk website.

```
wget -O splunk-enterprise.tgz  
'https://download.splunk.com/products/splunk/releases/9.2.0/linux/splunk-9.2.0-linux-2.6-amd64.tgz'
```

Step 2: Extract and install:

```
tar -xvzf splunk-enterprise.tgz -C /opt /opt/splunk/bin/splunk start --accept-license  
/opt/splunk/bin/splunk enable boot-start
```

Step 3: Configure receiving port for Universal Forwarder. Go to Settings > Forwarding and Receiving > Receive Data > Add New. Set port to 9997.

3.5.2 Splunk Universal Forwarder (Ubuntu Victim)

Install the Splunk Universal Forwarder on the Ubuntu victim machine to ship Ubuntu Event Logs to Splunk.

- Download the Universal Forwarder from splunk.com (wget link)
- Configure inputs.conf to monitor Ubuntu Event Logs:

```
[monitor:///var/log/apache2/access.log]  
index = web  
sourcetype = access_combined  
  
[monitor:///var/log/apache2/error.log]  
index = web  
sourcetype = error_log
```

3.5.3 Creating Splunk Alert for n8n

Create a saved search alert that fires when brute-force or suspicious login is detected:

```
index=web "/DVWA/vulnerabilities/brute/"
| rex field=_raw "(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| bin _time span=1m
| stats count as attempts by src_ip, _time
| where attempts > 3
```

Set the alert action to "Webhook" and point it to: <http://192.168.10.150:5678/webhook/splunk-alert>

Create a saved search alert that fires when SQLi or suspicious login is detected:

```
index=web "sql"
| rex field=_raw "(?<src_ip>\d+\.\d+\.\d+\.\d+)"
| eval Attack="SQL Injection"
| eval Attempts=1
| eval Severity="HIGH"
| eval Type="External Attack"
| table Attack, src_ip, Attempts, Severity, Type, _time
```

3.6 n8n Automation Engine (192.168.10.150)

3.6.1 System Requirements

- OS: Ubuntu Server 22.04 LTS
- RAM: 4 GB minimum
- CPU: 2 vCPUs
- Disk: 20 GB
- Port: 5678 (n8n Web UI and Webhook listener)

3.6.2 n8n Installation via Docker (Recommended)

Step 1: Install Docker and Docker Compose:

```
sudo apt update && sudo apt install -y docker.io docker-compose sudo systemctl enable --now docker
sudo usermod -aG docker $USER
```

Step 2: Create docker-compose.yml for n8n:

```
version: "3" services:  n8n:      image: n8nio/n8n      restart: always      ports:      -      "5678:5678"      environment:      - N8N_BASIC_AUTH_ACTIVE=true      - N8N_BASIC_AUTH_USER=admin      - N8N_BASIC_AUTH_PASSWORD=SOCpassword123      - WEBHOOK_URL=http://192.168.10.150:5678/      volumes:      - ~/.n8n:/home/node/.n8n
```

Step 3: Start n8n:

```
docker-compose up -d # Access UI at: http://192.168.10.150:5678
```

3.6.3 n8n Workflow Configuration

The core SOC automation workflow in n8n consists of the following nodes:

9. Webhook Node — Receives Splunk alerts at /webhook/splunk-alert
10. JSON Parse Node — Extracts alert fields (src_ip, user, event_count, timestamp)
11. AI Node (Gemini) — Calls Gemini API for threat enrichment
12. Slack Node — Posts formatted alert message to #soc-automation channel
13. Email Node — Sends incident email to SOC distribution list
14. SSH Node — Conditionally executes isolation/blocking commands on endpoint

3.7 AI Integration (Gemini)

3.7.1 Configuration

3.7.1 Step 1 — Get API Key

From Google AI Studio:

```
export GEMINI_API_KEY="AIzaSyXXXXXX"
```

3.7.3 Step 2 — Clean Prompt (IMPORTANT 🗨️)

Use a strong prompt so Gemini gives structured output:

You are a SOC analyst.

Analyze the log and respond ONLY in this format:

Attack: <SQL Injection / Brute Force / Legitimate>

Severity: <Low/Medium/High>

Reason: <short reason>

Log:

<your log here>

You can provide your own prompt.

3.7.4 Step 3 — Dynamic Log Input

Replace log manually with your real log:

Example:

```
"Log: 192.168.10.100 - - \"POST /DVWA/vulnerabilities/brute/ HTTP/1.1\""
```

3.8 Slack & Email Notifications

3.8.1 Slack Integration

Create a Slack app and obtain an Incoming Webhook URL from api.slack.com. In n8n, use the Slack node or HTTP Request node to POST formatted alert messages to your #soc-automation channel.

- Create a Slack workspace and #soc-automation channel
- Navigate to api.slack.com > Create App > Incoming Webhooks > Activate
- Copy the Webhook URL and configure it in the n8n Slack node
- Format the message payload with Slack Block Kit for rich alert formatting

3.8.2 Email Notifications

Configure the Email (SMTP) node in n8n to send incident notifications:

- Use Gmail SMTP: smtp.gmail.com, port 587, TLS enabled
- Generate an App Password from your Google account for authentication
- Configure n8n SMTP credentials: Settings > Credentials > Add SMTP
- Design email template with alert severity, IP address, username, and timestamp

3.9 SSH Response Action

3.9.1 Configuration

For automated response actions, n8n can execute SSH commands on the Ubuntu victim machine or other Linux endpoints. Configure the SSH node in n8n as follows:

- Enable SSH server on the target machine by selection options in pfsense
- Configure response actions: block IP via iptables, kill suspicious processes, quarantine files

```
pfctl -t blocked_ips -T add src_ip
```

This command should add in SSH node @ n8n

4. Automation Workflow

4.1 End-to-End Workflow Description

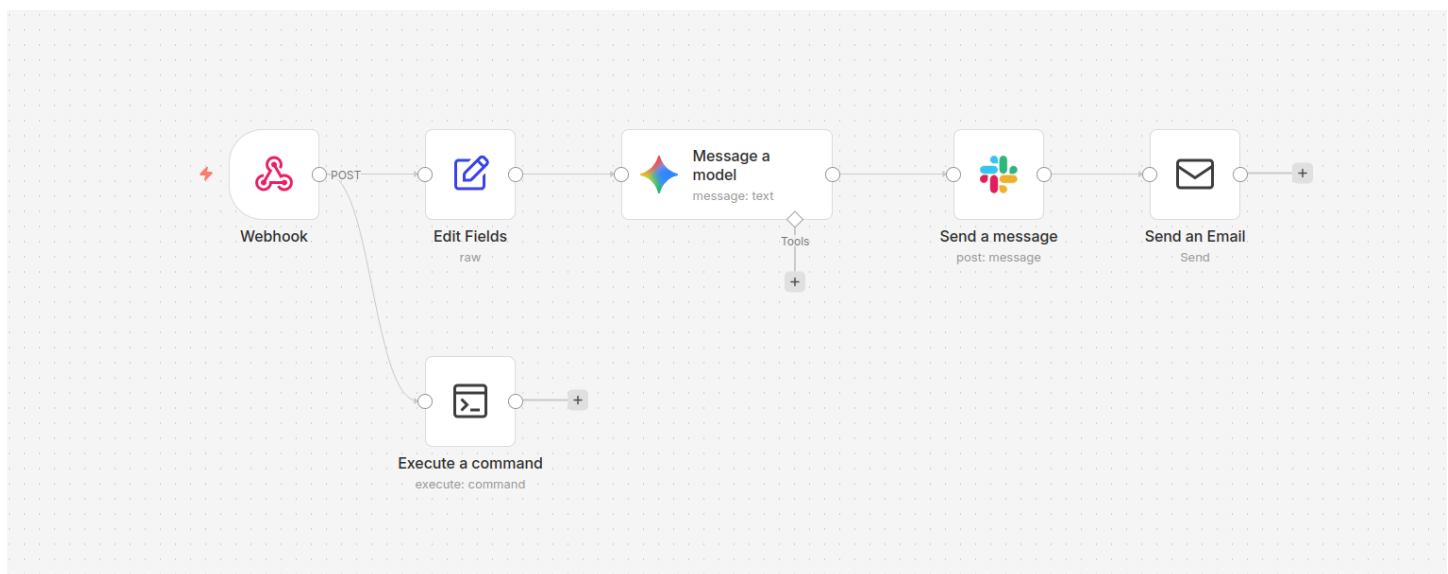
The complete SOC automation pipeline operates as follows when a threat is detected:

15. Attack Simulation: An external threat actor attempts brute-force login on the Ubuntu victim machine exposed to the internet.
16. Log Generation: In ubuntu events are generated and forwarded by Splunk Universal Forwarder to Splunk SIEM at 192.168.10.50.
17. Alert Trigger: Splunk's correlation search detects 3+ failed logins from the same source IP within 1 minutes and triggers a webhook alert to n8n.
18. Workflow Execution: n8n receives the webhook payload, parses the JSON data (src_ip, username, event_count, timestamp), and begins the automation workflow.
19. AI Enrichment: The alert is sent to the Gemini. The AI returns a threat classification, recommended severity level, and suggested response actions.
20. Analyst Notification: Slack message is posted to #soc-automation.
21. Email Dispatch: An incident email is sent to the SOC team distribution list.
22. Automated Response: If AI classifies severity as HIGH, n8n executes SSH commands on the endpoint to block the attacker IP via iptables.

4.2 n8n JSON Template

Import this to JSON node (n8n)

```
{ "Attack": "={{ $json.body.result.Attack || ($json.body.search_name === 'bruteforceattack' ? 'Brute Force' : 'SQL Injection') }}", "Attacker": "={{ $json.body.result.src_ip }}", "Attempts": "={{ $json.body.result.Attempts || $json.body.result.attempts || '1' }}", "Time": "={{ new Date($json.body.result._time * 1000).toLocaleString() }}", "Severity": "={{ $json.body.result.Severity || ($json.body.search_name === 'bruteforceattack' ? 'MEDIUM' : 'HIGH') }}", "Type": "={{ $json.body.result.Type || 'External Attack' }}" }
```



5. Testing & Validation

5.1 Test Cases

#	Test Case	Expected Result	Status
1	Brute-force login simulation on Ubuntu victim	Splunk alert fires, n8n workflow triggered	Successful
2	Splunk webhook delivery to n8n	n8n receives JSON payload successfully	Successful
3	AI enrichment API call	Gemini returns threat classification and severity	Successful
5	Slack notification delivery	Alert posted to #soc-automation within 30 seconds	Successful
6	Email notification	Incident email received by SOC team	Successful
7	SSH automated response	Attacker IP blocked via iptables on endpoint	Successful

6. Security Considerations

6.1 Lab Network Hardening

- Isolate the victim Ubuntu machine using a dedicated VLAN or firewall rules to prevent lateral movement to production networks
- Use strong, unique passwords for all tool admin interfaces (Splunk, n8n)
- Rotate API keys regularly and store them as environment variables, never hardcoded
- Restrict n8n webhook endpoint access to Splunk server IP only using iptables
- Enable TLS/HTTPS for all tool web interfaces using self-signed or Let's Encrypt certificates
- Limit SSH access to the automation engine using key-based authentication only

6.2 Data Privacy

- Do not use real personal data in the lab environment
- Sanitize logs before sharing externally or including in documentation
- Review AI API terms of service regarding data sent to external APIs

7. Troubleshooting

7.1 Common Issues

Issue	Possible Cause	Resolution
Splunk not receiving forwarder data	Port 9997 blocked or forwarder misconfigured	Check iptables, verify outputs.conf on forwarder
n8n webhook not triggering	Wrong webhook URL in Splunk alert	Verify URL: http://192.168.10.150:5678/webhook/splunk-alert
AI API returns error	Invalid API key or rate limit exceeded	Gemini key validity and account limits
Slack notification not received	Webhook URL expired or channel deleted	Regenerate Slack incoming webhook URL
SSH connection refused	SSH not enabled or firewall blocking port 22	Enable SSH service and open port 22 in firewall

8. Attack Scenario & Workflow Execution

Overview

This section explains how a real-world attack is simulated in the SOC-Automation-Lab and how the system detects, analyzes, and responds automatically.

The attack is performed on the Ubuntu victim machine (192.168.10.20) hosting the DVWA application.

8.1 Attack Scenario 1: Brute Force Attack

Step 1: Attack Execution

An attacker machine (Kali Linux) sends multiple login requests to:
[/DVWA/vulnerabilities/brute/](#)

Vulnerability: Brute Force

Login

Username:

Password:

Username and/or password incorrect.

Step 2: Log Generation

Apache logs the repeated login attempts in:
`/var/log/apache2/access.log`

Step 3: Log Forwarding

Step 4: Detection (Splunk)

Triggered Alerts

Filter App Search & Report... Owner All owners Severity All severity Alert name All alerts Showing 1 - 1 of 1 results Show: 100 per page 1 of 1 pages

☐	Time	Alert name	App	Type	Severity	Mode	Actions
☐	2026-04-29 17:42:36 UTC	bruteforceattack	search	Real-time	High	Per Result	View Results Edit Alert Delete

Step 5: Alert to n8n

<http://192.168.10.150:5678/webhook/splunk-alert>

Webhook

Pull in events from Webhook

Listen for test event

Once you've finished building your workflow, run it without having to click this button by using the production webhook URL. [More info](#)

ParametersSettingsListen for test event

Webhook URLs

Test URLProduction URL

POSThttp://localhost:5678/webhook-test/sql-alert

HTTP Method
POST

Path
sql-alert

Authentication
None

Respond
Immediately

If you are sending back a response, add a "Content-Type" response header with the appropriate value to avoid unexpected behavior

Options
No propertiesAdd option

OUTPUT

headersaccept-encodingidentitycontent-length419host192.168.10.100:5678content-typeapplication/jsonuser-agentSplunk/C307CCD-FD7C-4C97-9C25-93D33F8E9F4connectioncloseparamsquerybody{"sid":"rt_scheduler_bhuvi_search_bruteforceattack_at_1777536900_722","search_name":"bruteforceattack","app":"search","owner":"bhuvi","results_link":"http://splunk:8000/app/search/search?q%3Dhead%3Drt_scheduler_bhuvi_search_bruteforceattack_at_1777536900_722%20%7C%20head%20%7C%20tail%20&earliest=0&latest=now"}result{"src_ip":"192.168.10.100","time":1777537140,"attempts":4}webhookUrlhttp://localhost:5678/webhook-test/sql-alertexecutionModetest

Step 6: AI Analysis (Gemini)

Message a model

Schema Table JSON

Parameters Settings

Execute step

INPUT

1 item

Attack

Brute Force

Attacker

192.168.10.100

Attempts

4

Time

4/30/2026, 8:19:00 AM

Severity

MEDIUM

Type

External Attack

Webhook

1 item

Variables and context

Model

From list

models/gemini-3-flash-preview

Messages

Prompt

You are a professional SOC (Security Operations Center) Analyst.
Analyze the following security event and generate a clear, structured incident report.

Role

User

Add Message

Simplify Output

Output Content as JSON

OUTPUT

Schema Table JSON

1 item

content

parts

parts[0]

text

++INCIDENT ANALYSIS REPORT++
++Threat Level++
++Low++
(Note: While Brute Force is a serious method, the low volume of attempts (4) suggests either a manual probe or an early-stage automated scan rather than an ongoing massive breach.)
++Attack Type++
++Brute Force (Credential Guessing)++
++Source++
++Attacker IP++ 192.168.10.100
++Attack Category++ External Attack
++Technical Analysis++ The system detected a series of 4 failed authentication attempts originating from an external source (192.168.10.100). In a Brute Force attack, a threat actor attempts to gain access...

thoughtSignature

EpKwCpYWAQw5tfaX6sPB4cz2mTrchH96ioOfpmsLbCoTRBoaBnD25Z3bYVhYkGndCphtguU7KOAYz7o7ggCvHlUln4L1ZJqAJU8DQZFY0GeCAZYX8uzsvLk9dEUnkAHwpr85d6nTHL135NmqaH9Fmrv30V50InNnCa+Roeml+mH+H2W3qkAdpYz2gAdCUUv3k+eogynoZxd7HBL5fWV9W25TovmBfQJNB0hU7Y8kSLEqu3Bk55fQOMQVBLmNHNtXp3SHG5B6Q4m7a0hHq0XWYnyfQKYNhuw8Q2ZVL0C4d8mUu-wfM8t0N0mporyP7aDae3kU0D0wghhemp1v9SL0g0JmgsEVAuxRrw0HBPZ1v7aImuUo+hS0dZ2AP4HDS/89d2UuLg7fJ8MkZyruUMFSp0c0SHvhs35aTmpeUeA7fP0H+59vygvrFMKcyJ97LVB30cBTY1Aox8Eeo0Bohm7vP4RqduMR66wA3Y0utsv1G5xZz30Dq3JlqCFM+8kL7NUTUusFchn25M+kk...

role

model

finishReason

STOP

index

0

Step 7: Notification

Today

++INCIDENT ANALYSIS REPORT++

++Threat Level++
++Low++
(Note: While Brute Force is a serious method, the low number of attempts (4) suggests this may be an initial probing phase or an automated bot. However, it requires immediate attention to prevent escalation.)

++Attack Type++
++Brute Force Attack++

++Source++
++Attacker IP++ 192.168.10.100
++Attack Category++ External Attack

++Technical Analysis++
A Brute Force attack occurs when an actor attempts to gain unauthorized access to a system by systematically trying multiple usernames and passwords until the correct combination is found. In this instance, our monitoring tools flagged ++4 failed login attempts++ in quick succession from the external IP ++192.168.10.100++. This pattern is consistent with "credential probing," where an attacker tests common passwords against known accounts.

++Impact++
++Unauthorized Access++ If successful, the attacker could gain control over a user account, leading to data theft or further network penetration.
++Account Lockout++ Repeated failed attempts may trigger security policies that lock legitimate users out of their accounts, causing a disruption in business operations.
++System Integrity++ An intruder could potentially install malware or modify system configurations once inside.

++Mitigation++
++IP Blocking++ Temporarily block or blacklist the source IP (192.168.10.100) at the firewall level.
++Account Lockout Policy++ Ensure a policy is active that locks accounts after a specific number of failed attempts (e.g., 5 attempts).
++Enable MFA++ Enforce Multi-Factor Authentication (MFA) to ensure that even if a password is guessed, the attacker cannot gain access.
++Log Monitoring++ Continue monitoring logs for this IP to see if it targets other services or changes its tactics.

++Final Verdict++
The event is currently classified as a ++low-level probing attempt++. No successful breach has been detected. By implementing the suggested mitigations, the risk of a successful compromise from this source remains minimal. High-priority monitoring is recommended for the next 24 hours.

Automated with this n8n workflow


bhuvankul4u@gmail.com
 to bhuvancyberspace

2:43 PM (0 minutes ago) ☆ ☺ ↶ ⋮

:shield: ****INCIDENT ANALYSIS REPORT****

****red_circle: Threat Level:****
****Low****
 (Note: While Brute Force is a serious method, the low number of attempts (4) suggests this may be an initial probing phase or an automated bot. However, it requires immediate attention to prevent escalation.)

****dart: Attack Type:****
****Brute Force Attack****

****round_pushpin: Source:****
 * ****Attacker IP:**** 192.168.10.100
 * ****Attack Category:**** External Attack

****brain: Technical Analysis:****
 A Brute Force attack occurs when an actor attempts to gain unauthorized access to a system by systematically trying multiple usernames and passwords until the correct combination is found. In this instance, our monitoring tools flagged ****4 failed login attempts**** in quick succession from the external IP ****192.168.10.100****. This pattern is consistent with "credential probing," where an attacker tests common passwords against known accounts.

****warning: Impact:****
 * ****Unauthorized Access:**** If successful, the attacker could gain control over a user account, leading to data theft or further network penetration.
 * ****Account Lockout Policy:**** Repeated failed attempts may trigger security policies that lock legitimate users out of their accounts, causing a disruption in business operations.
 * ****System Integrity:**** An intruder could potentially install malware or modify system configurations once inside.

****hammer_and_wrench: Mitigation:****
 * ****IP Blocking:**** Temporarily block or blacklist the source IP (192.168.10.100) at the firewall level.
 * ****Account Lockout Policy:**** Ensure a policy is active that locks accounts after a specific number of failed attempts (e.g., 5 attempts).
 * ****Enable MFA:**** Enforce Multi-Factor Authentication (MFA) to ensure that even if a password is guessed, the attacker cannot gain access.
 * ****Log Monitoring:**** Continue monitoring logs for this IP to see if it targets other services or changes its tactics.

****scroll: Final Verdict:****
 The event is currently classified as a ****low-level probing attempt****. No successful breach has been detected. By implementing the suggested mitigations, the risk of a successful compromise from this source remains minimal. High-priority monitoring is recommended for the next 24 hours.

Step 8: Automated Response

: pfctl -t blocked_ips -T add <src_ip>

“Attacker IP is blocked”

```

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
                v6/DHCP6: fd17:625c:f037:2:a00:27ff:fee5:448/64
LAN (lan) -> em1 -> v4: 192.168.10.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T show
192.168.10.20
[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T delete 192
.168.10.20
1/1 addresses deleted.
[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T show
[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T show
192.168.10.100
[2.8.1-RELEASE][root@pfSense.home.arpal/root:

```

8.2 Attack Scenario 2: SQL Injection Attack

Step 1: Attack Execution

```
' OR '1'='1
```



Step 2: Log Generation

Step 3: Detection (Splunk)

Triggered Alerts

Filter	App	Search & Report...	Owner	All owners	Severity	All severity	Alert name	All alerts	Showing 1 - 2 of 2 results	Show: 100 per page	1	of 1 pages
☐	Time											
☐	2026-04-29 09:09:11 UTC		bruteforceattack		search	Real-time	High	Per Result	View Results	Edit Alert	Delete	
☐	2026-04-29 09:09:55 UTC		sqlalert		search	Real-time	High	Per Result	View Results	Edit Alert	Delete	

Step 4: Alert to n8n

Webhook

Pull in events from Webhook

[Listen for test event](#)

Once you've finished building your workflow, run it without having to click this button by using the production webhook URL. [More info](#)

Parameters

Webhook URLs

Test URL Production URL

POST http://localhost:5678/webhook-test/sql-alert

HTTP Method

POST

Path

sql-alert

Authentication

None

Respond

Immediately

If you are sending back a response, add a "Content-Type" response header with the appropriate value to avoid unexpected behavior

Options

No properties

[Add option](#)

OUTPUT

1 item

headers

accept-encoding identity

content-length 472

host 192.168.10.150:5678

content-type application/json

user-agent Splunk/C3077CCD-FD7C-4C97-9C25-93033FB8E9F4

connection close

params

query

body

sid rt_scheduled_bhuvi_search_sqlaler_at_3777531080_7070

search_name sqlalert

app search

owner bhuvi

results_link http://splunk:8000/app/search/search?q=%7Cload%20rt_scheduled_bhuvi_search_sqlaler_at_3777531080_7070%20head%20tail%20%7C%20tail%20%earliest=0klatest=now

result

Attack SQL Injection

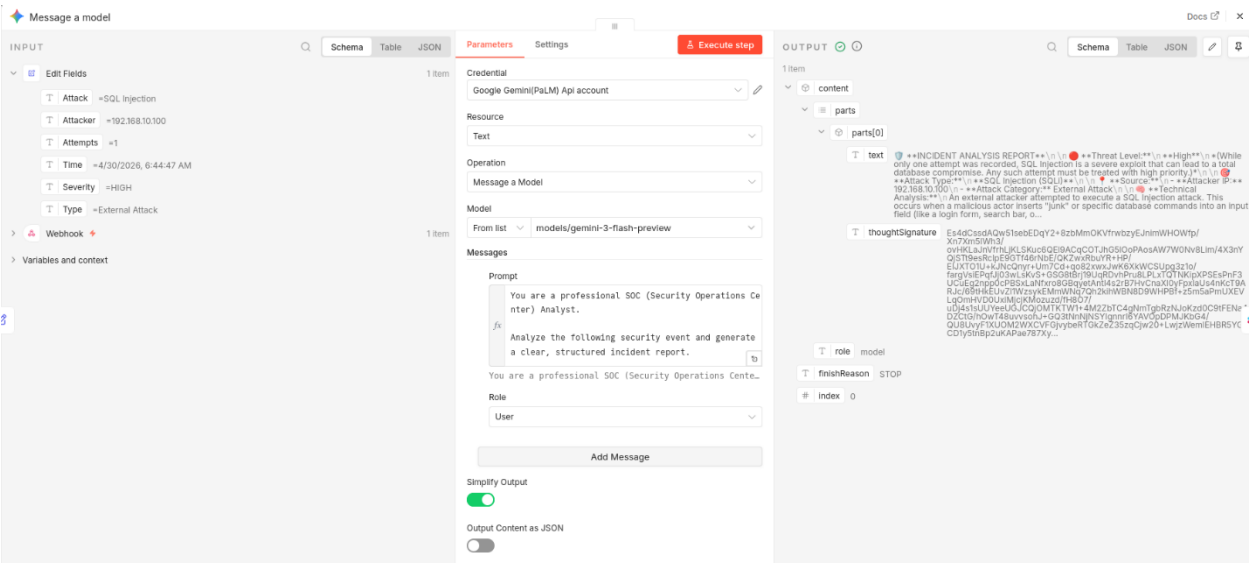
src_ip 192.168.10.100

Attempts 1

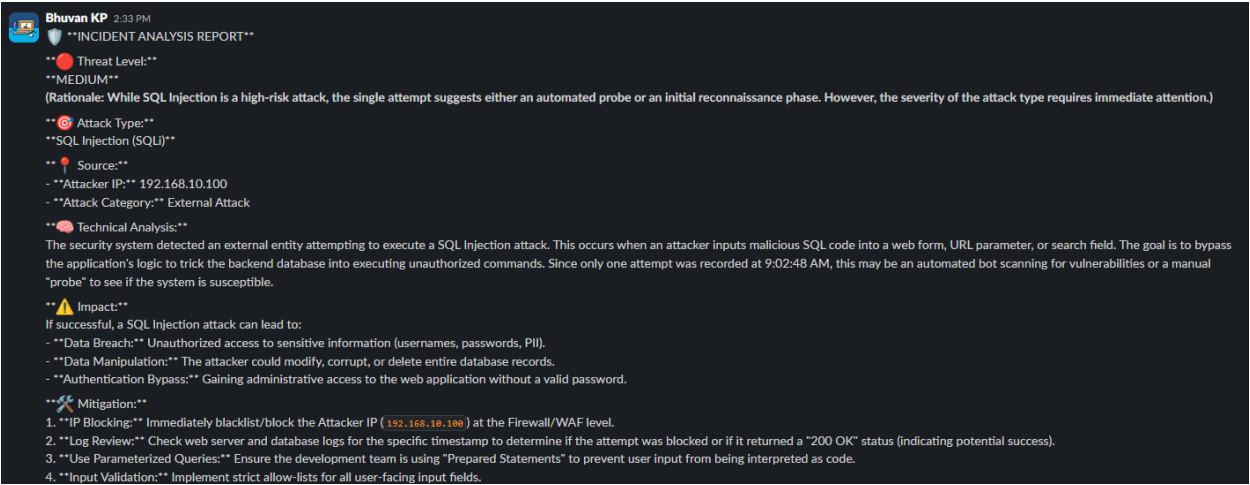
Severity HIGH

Type External Attack

Step 5: AI Enrichment



Step 6: Notification





bhuvankulal4u@gmail.com

to me ▾

2:33 PM (3 minutes ago)



:shield: **INCIDENT ANALYSIS REPORT**

red_circle: Threat Level:

MEDIUM

(Rationale: While SQL Injection is a high-risk attack, the single attempt suggests either an automated probe or an initial reconnaissance phase. However, the severity of the attack type requires immediate attention.)

dart: Attack Type:

SQL Injection (SQLi)

round_pushpin: Source:

- **Attacker IP:** 192.168.10.100

- **Attack Category:** External Attack

brain: Technical Analysis:

The security system detected an external entity attempting to execute a SQL Injection attack. This occurs when an attacker inputs malicious SQL code into a web form, URL parameter, or search field. The goal is to bypass the application's logic to trick the backend database into executing unauthorized commands. Since only one attempt was recorded at 9:02:48 AM, this may be an automated bot scanning for vulnerabilities or a manual "probe" to see if the system is susceptible.

warning: Impact:

If successful, a SQL Injection attack can lead to:

- **Data Breach:** Unauthorized access to sensitive information (usernames, passwords, PII).
- **Data Manipulation:** The attacker could modify, corrupt, or delete entire database records.
- **Authentication Bypass:** Gaining administrative access to the web application without a valid password.

hammer_and_wrench: Mitigation:

1. **IP Blocking:** Immediately blacklist/block the Attacker IP ('192.168.10.100') at the Firewall/WAF level.
2. **Log Review:** Check web server and database logs for the specific timestamp to determine if the attempt was blocked or if it returned a "200 OK" status (indicating potential success).

Step 7: Automated Response

```

WAN (wan) -> em0 -> v4/DHCP4: 10.0.2.15/24
                v6/DHCP6: fd17:625c:f037:2:a00:27ff:fee5:448/64
LAN (lan) -> em1 -> v4: 192.168.10.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T show
192.168.10.20
[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T delete 192
.168.10.20
1/1 addresses deleted.
[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T show
[2.8.1-RELEASE][root@pfSense.home.arpal/root: pfctl -t blocked_ips -T show
192.168.10.100
[2.8.1-RELEASE][root@pfSense.home.arpal/root:

```

9. End-to-End Flow Summary

10. Future Enhancements

- Integrate Cortex for automated threat intelligence enrichment (VirusTotal, Shodan, AbuseIPDB)
- Implement MITRE ATT&CK mapping to automatically tag alerts with tactic and technique IDs
- Add more attack simulations: Ransomware, C2 communication, data exfiltration scenarios
- Implement automated vulnerability scanning with OpenVAS/Greenbone triggered post-incident
- Explore LLM-based playbook generation — allow the AI to suggest and create n8n workflows
- Deploy in cloud (AWS/Azure) for a hybrid SOC lab experience

11. References & Resources

Official Documentation

- Splunk Documentation: <https://docs.splunk.com>
- n8n Documentation: <https://docs.n8n.io>
- Gemini API: <https://docs.anthropic.com>
- ChatGPT
- Claude AI